MASTER

Player Types and Experience Dimensions

Modelling Player Experience for Achievers and Philanthropists in a Gamified Cognitive Task

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Player Types and Experience Dimensions:
Modelling Player Experience for Achievers and Philanthropists
in a Gamified Cognitive Task

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Abstract

This thesis explores how Hexad player types, specifically Achievers and Philanthropists, relate to player enjoyment, and how this relationship is shaped by specific player experience dimensions measured via the Player Experience Inventory (PXI). Using a large-scale dataset (N = 1,094) from the cognitive game “Tunnel Runner”, a series of mediation and moderation analyses were conducted. Model analyses revealed that player experience dimensions mediate this relationship and operate through sequential pathways. For Achievers, enjoyment is primarily driven by mastery, and challenge, facilitated by clarity of goals and rules, and progress feedback. For Philanthropists, enjoyment is rooted in meaning, which is influenced by progress feedback and clarity of goals and rules. These findings emphasize that enjoyment in gamified environments arises not from isolated elements, but from interconnected experiences that align with players’ motivational profiles. The study highlights the importance of designing for these sequential experiential pathways to better engage diverse user types in gamified systems.

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1. Introduction

Games were originally designed with the primary goal to entertain and engage users, and this idea is used in gamification to increase fun and engagement in other goods and services (Deterding et al., 2011). Gamification is the application of game design elements in non-game contexts (Deterding et al., 2011), and has enhanced engagement across different fields, e.g., education (Zeybek & Saygı, 2024), healthcare (Hopia & Raitio, 2016), and the workplace (Dale, 2014). However, individual experiences with gamification can vary significantly (Koivisto & Hamari, 2014). As a result, gamification has shifted from a one-size-fits-all approach to more tailored strategies (Klock et al., 2020). It is known that different players have different motivational drivers (Lawrence et al., 2002), and tailored gamification uses this knowledge to adapt game elements to individual needs (Klock et al., 2020).

One widely used method for tailored gamification is Hexad, a player-type framework specifically developed for gamification. Hexad defines six gamification user types: Achiever, Free Spirit, Philanthropist, Socializer, Player, and Disruptor (Marczewski, 2015), and is based on insights from Self-Determination Theory (SDT; Deci & Ryan, 1985). Hexad types are assumed to have different motivational focuses when playing games, influencing how users interact with gamified systems and experience enjoyment (Marczewski, 2015). For example, Achievers are motivated by competence and challenge, and they thrive in environments that reward their achievements and progress. Philanthropists are driven by altruism and purpose, helping others and contributing to the greater good. Research has shown the relationship between Hexad player types and preferences for specific game elements (Krath & von Korflesch, 2021), e.g., Achievers prefer elements such as points, badges, and leaderboards, as they provide visual feedback on their progress and performance. The varying preferences of different player types are often used to personalize gamified systems. This tailoring process aims to align the system's features with the user's dominant player type to better support their core motivations.

While many studies have explored the relationship between Hexad types and game element preferences (Krath & von Korflesch, 2021; Tondello et al., 2016), less is known about how player types experience a game differently. Hamari et al. (2014) highlighted that findings on the effectiveness of tailored gamification in increasing player motivation and performance are often contradicting. Some studies find positive effects of tailored compared to non-tailored systems (e.g. Hallifax et al., 2020; Serna et al., 2023), while others do not (e.g. Lavoué et al., 2019; Oliveira et al., 2022). This might indicate that there are other factors that influence players' motivation, rather than just the presence of game elements themselves. The contradicting results raise questions about how different players experience gamification differently. Researchers have developed several methods to measure player experience, e.g., the Game Experience Questionnaire (GEQ; IJsselsteijn et al., 2007), Player Experience of Needs Satisfaction (PENS; Ryan et al., 2006), User Engagement Scale (UES; O'Brien et al., 2018), and Player Experience Inventory (PXI; Vanden Abeele et al., 2020). These validated scales either reflect specific theories or address isolated constructs. The PXI is particularly useful as it includes a broad range of player experience dimensions.

The Player Experience Inventory (PXI; Vanden Abeele et al., 2020) integrates functional and psychological dimensions and aims to integrate a broad range of theories and concepts such as Self-Determination Theory (Deci & Ryan, 1985) and Means-End Theory (Gutman, 1982). The PXI is based on the idea that game design choices, such as interface features or specific game mechanics, shape how players perceive and interact with a gamified system. These design choices result in functional consequences, including *ease of use, progress feedback, audiovisual appeal, clarity of goals and rules,* and *challenge*. These are the direct and observable effects of game design choices. In contrast, psychological consequences, including *mastery, curiosity, immersion, autonomy*, and *meaning*, reflect deeper emotional and cognitive responses to gameplay. Together, the PXI offers a structured way to evaluate how these elements contribute to player enjoyment.

Since gamification aims to make activities more enjoyable and engaging (Deterding et al., 2011), understanding what contributes to enjoyment is particularly valuable. In this thesis, enjoyment is used as the main outcome to explore how different Hexad player types experience a gamified system differently. I will focus on understanding players' unique experiences through the Player Experience Inventory (PXI). The study draws from recently published data collected in an online study (N=1094) conducted by Markovitch et al. (2025). The dataset includes Hexad player type scores and PXI responses based on participants' experiences with the gamified cognitive assessment "Tunnel Runner" (Markovitch et al., 2024). Tunnel Runner is an infinite runner game that has been designed to measure cognitive processes, including response inhibition (Diamond, 2013), interference control (Diamond, 2013), and response-rule switching (Schroder et al., 2012).

This thesis focuses on two specific Hexad types: Achievers— driven by competence and mastery, focusing on personal growth and success—and Philanthropists—motivated by relatedness and purpose, with a stronger emphasis on helping others. These were selected because 1) they are the most prominent player types in previous research; 2) they are among the most prominent player types in our data set; and 3) they reflect clearly contrasting motivational profiles enabling meaningful comparison. Gaining more insight into these specific player types is useful, because they align with the most commonly used gamification elements, such as points, leaderboards, badges, social features, and teams (Hamari et al., 2014; Klock et al., 2020; Koivisto & Hamari, 2019). Therefore, they are more likely to engage positively with many gamified systems, making them a relevant starting point for investigating how different types of player experiences shape enjoyment.

To explore how Achievers and Philanthropists experience gamified systems, I focus on the PXI dimensions theoretically most relevant to these two player types. For Achievers, relevant dimensions include *mastery, challenge, progress feedback,* and *clarity of goals and rules*. These dimensions reflect their motivation to develop skills and tackle challenges, which require clear feedback on their performance and well-defined goals. For Philanthropists, *meaning, progress feedback* and *clarity of goals and rules* are most relevant, as they align with a drive for purpose and helping others: *meaning* reflects their desire to contribute to something larger than themselves; *progress feedback* helps them understand how their actions benefit others and track their impact; and *clarity of goals and rules* ensures they can understand how to contribute in a meaningful way within the system. The selection of these experience dimensions is in line with psychological theories such as Self-Determination Theory (Deci & Ryan, 1985), Uses and Gratifications Theory (Katz et al., 1973), Motive Disposition Theory (Deterding et al., 2011; Klock et al., 2020), and the Four Drives Theory (Lawrence et al., 2002).

The central research question in this thesis is: *How do differences in player experience explain the relationship between player type scores and enjoyment?* To answer the central research question, I investigate whether PXI dimensions mediate, moderate, or interact in more complex ways in the relationship between player type and enjoyment. By systematically testing and comparing these competing models, the goal is to identify the theoretical framework that best explains the relationship between player type scores and enjoyment. Understanding how individual differences, i.e. Hexad types, lead to different game experiences adds to the field of understanding player experience. Furthermore, this study aims to offer practical insights for game developers that can be used to develop more personalized experiences that better suit the motivations of different player types.

My findings demonstrate that Achievers find enjoyment primarily through *challenge* and *mastery*, while Philanthropists are driven by *meaning*. Interestingly, the models show sequential pathways, demonstrating that player experience dimensions build on each other to shape enjoyment, rather than operating purely in isolation. These insights suggest that gamified systems are more effective when they are built around interconnected experience dimensions, rather than relying on isolated elements.

1.1 Thesis Structure

The remainder of my thesis is structured as follows: Chapter 2 “Related Work” reviews the existing literature on gamification, player typologies, and the Player Experience Inventory (PXI), which forms the theoretical basis for this study. Chapter 3 “Methods” outlines the methods used, including a description of Tunnel Runner, participant details, measurements, and data analysis approach. Chapter 4 “Results” presents the results of the analyses. Structural equation modeling (SEM) was used to test a series of mediation and moderation models to examine how selected PXI dimensions explain the relationship between player type scores and enjoyment. Based on these analyses, final models were developed separately for both player type profiles. Chapter 5 “Discussion” discusses these findings and their implications for gamification design, along with the study’s limitations and suggestions for future research. Chapter 6 “Conclusion” concludes the thesis.

2. Related Work

This section will discuss the theoretical background of gamification, player typologies, and player experience inventory. First, relevant literature on gamification will be discussed to allow readers unfamiliar with this topic to understand what it is and how research in this field evolved. Then, I will discuss several relevant player typologies that have been developed over the years, and a detailed explanation of the Hexad model. This will allow readers to better understand the context related to player typologies and how the different models differ from each other. Moreover, the Hexad model will be connected to several psychological theories to explain how psychological theories relate to the underlying motivations of Achievers and Philanthropists. Lastly, the PXI model will be explained in more detail, along with studies that have applied it in past research.

2.1 Gamification

Gamification is commonly defined as the use of game elements in non-game contexts (Deterding et al., 2011). Rather than the application being gameful itself, these elements should allow gameful interactions within the system (Deterding et al., 2011). The early research on gamification at the beginning of the 2010s showed that gamified systems could boost motivation, engagement, and improve enjoyment (Hamari et al., 2014). These studies were mainly focused on reward-based mechanics, such as points, badges, leaderboards, and levels. Yet, over time, research emphasized the importance of higher-level game mechanics, such as storytelling, social interaction, and stimuli for creativity (Fischer et al., 2016). Today, gamification has been widely implemented in various fields such as education (Zeybek & Saygı, 2024), healthcare (Hopia & Raitio, 2016), and the workplace (Dale, 2014), shaping user behavior in more meaningful and sustainable ways.

Most gamification research studies a gamified system that includes multiple gamification elements together, but the effects of individual gamification elements have also been studied separately. For example, Mekler et al. (2017) studied the effect of multiple individual gamification elements on performance, intrinsic motivation, and need satisfaction. In their study, they found that points, leaderboards, and badges individually increase performance. However, against their expectations, no effects have been found of the individual elements on intrinsic motivation and need satisfaction.

Recently, gamification has shifted from the one-size-fits-all approach to more tailored strategies that adapt game elements to individual needs (Klock et al., 2020). As individual differences in the effects of gamification were observed, this raised questions about whether different gamification approaches should be used for different people. It is known that different players have different motivational drivers (Lawrence et al., 2002), and this has been used as one of the explanations for personalized or tailored gamification. By aligning game elements with players' psychological needs for autonomy, competence, and relatedness, it is possible to enhance intrinsic motivation and, in turn, player enjoyment (Mekler et al., 2017). Gamification can be tailored to each user's preferences by generating user profiles to categorize players. The majority of these profiles are based on personality traits, gender, or player type (Hallifax et al., 2020). Based on these profiles, different gamification elements are recommended in order to increase motivation, engagement, and performance.

To better understand the effectiveness of tailored gamification approaches, several studies have examined whether matching game elements to player types leads to improved outcomes such as enjoyment, motivation, and engagement. Oliveira et al. (2022) examined whether a personalized gamification environment based on player type affects students' experience of flow, enjoyment, and motivation. At the start of the experiment, participants completed the BrainHex (Nacke et al., 2014) questionnaire to identify the students' player types. Based on player type, a different combination of game elements was present in the gamified system. The researchers were not able to find any significant results for the experience of flow or motivation. Surprisingly, the results suggested that the Achiever and Socializer types enjoyed the non-personalized environment better.

Moreover, Lavoué et al. (2019) investigated the impact of tailored gamification on motivation, participation, and enjoyment in a real-world learning environment. Participants were divided into three groups: the adaptive group received features tailored to their BrainHex player profile (Nacke et al., 2014), the counter-adaptive groups received the least-matching features, and the control group had no gamified elements. Results indicated that the adaptive features group showed higher levels of participation and lower levels of amotivation. However, there was no significant difference in enjoyment between the groups. This suggests that enjoyment does not depend on a player's motivation and participation.

Other research on tailored gamification also focused on personalization based on goal alignment. In a different approach to tailored gamification, Leung et al. (2023) investigated how modifying gamified performance feedback to align with learners' goal orientations affected their engagement and performance in an online learning environment. The researchers customized feedback by varying the type of comparison (social vs. personal) and message framing (positive vs. negative) based on individual goal orientations. The study found that aligning gamified feedback with learners' specific goal orientations significantly enhanced their engagement and academic performance and highlighted the importance of personalized feedback in educational gamification.

While much of the research on tailored gamification has focused on individual differences in player types or goal orientations, demographic characteristics such as age and gender also play a key role in how people respond to gamified systems. Earlier studies on technology show that younger users place a higher importance on the usefulness of technology, whereas older generations are more influenced by social factors (Morris & Venkatesh, 2000). The digital divide is an explanation for this difference between generations. Younger users of technology have been exposed to technology already early on in their lives and this affects their attitude towards new technologies (Donat et al., 2009). These effects have also been observed in practice. Younger adults are more motivated by competition and performance, while older adults prefer cooperative and goal-oriented mechanics focused on completion and enjoyment (Birk et al., 2017). Gender differences have also been observed, with studies indicating that women tend to engage more with social and collaborative elements (Koivisto & Hamari, 2014), while men are more likely to be achievement-oriented and are more attracted to competition, rankings, and challenge-based mechanics (Williams et al., 2009).

The work on tailored gamification shows the importance of the specific individual differences a system is tailored to. One of the most common approaches to personalization involves player typologies, such as Bartle's taxonomy (Bartle, 1996), BrainHex (Nacke et al., 2014), and Hexad (Marczewski, 2015). Each typology has a different theoretical background and set of assumptions about how players differentiate in their experience, making it important to understand the theory in context. In the next section, I will describe different player typologies that have been developed over time, ending with a detailed explanation of the Hexad model, as this study focuses on Hexad player types.

2.2 Player Typology

As gamification research has evolved, it has become clear that different players are motivated by different game elements. While some players are driven by competition and mastery, others prefer social interactions or meaningful experiences. Recognizing these differences, researchers have categorized players into various typologies to better understand how they engage with gamified systems. In this section, different player typologies will be discussed and compared.

2.2.1 Player typologies over time

One of the earliest and most influential player type models is Bartle's taxonomy, originally developed for multiplayer games (Bartle, 1996). He identified four main motivations for playing Multi-User Dungeons (MUDs) games, namely achievements in the game, exploration of the game, socializing, and imposing upon others. Based on these motivations he identified four player types; Achiever, Explorer, Socializer, and Killer. However, there is some critique of this model as the player types are limited to a certain gameplay and cannot be used universally (Tondello et al., 2016). Therefore, other researchers have put forth more comprehensive typologies.

Following earlier player typologies like Bartle's, the Demographic Game Design (DGD) models (Bateman & Boon, 2005) offer a broader view of player types by integrating concepts from the Myers-Briggs Type Indicator (MBTI) (Myers, 1962). DGD1 introduces the four player archetypes Conqueror, Manager, Wanderer, and Participant, each reflecting different gameplay preferences and cognitive styles. The DGD2 builds on this by considering additional dimensions, such as the spectrum from casual to hardcore players, skill level variation, and preferences for solo or multiplayer experiences (Bateman et al., 2011).

In 2006, Yee (2006) identified five motivations for playing Massively Multiplayer Online Role-Playing Games (MMORPGs), based on Bartle's typology. However, Yee used a more data-driven approach to classify different motivations of players. These motivations are Achievement, Relationship, Manipulation, Immersion, and Escapism. The first three motivations, Achievement, Relationship, and Manipulations, are in line with Bartle's Achiever, Socializer, and Killer player types. Yee discovered the additional motivations Immersion and Escapism in his study. Immersion relates to how much users are immersed in the fantasy world, narrative, or role-playing of the game. Escapism describes how gamers use the virtual world to get away from their duties or troubles in real life.

Later, many different player type models have been proposed that focus on specific users. For example, Xu et al. (2012) developed a player typology based on the evaluation of a health game for young adults. This model does not only include motivational factors, but also includes behavioral influences. The resulting player types are Achievers, Active Buddies, Social Experience Seekers, Team Players, and Freeloaders.

Another very popular player typology is the BrainHex model (Nacke et al., 2014). The BrainHex model builds on earlier player typologies and incorporates insights from neurobiological research to categorize players into seven player types: Achiever, Conqueror, Daredevil, Mastermind, Seeker, Socializer, and Survivor. This model links the human body's neurobiological responses to the motivation of players. As each type includes contrasting styles, this model also helps highlight what a player might enjoy less (Barata et al., 2014). BrainHex has been widely implemented in Human-Computer Interaction (HCI) studies because of its diverse character and user-friendly online questionnaire.

While these presented models are very useful, a critique is that they were built specifically for game design (Tondello et al., 2016). This makes their application in gamified context limited. To address this gap, the Hexad framework was introduced, offering player types specifically tailored to the principles of gamification (Marczewski, 2015; Tondello et al., 2016). Unlike other models, the Hexad framework defines its user types as representations of players' intrinsic and extrinsic motivations, based on the Self-Determination Theory (SDT) (Deci & Ryan, 1985). The SDT describes three types of intrinsic motivation, namely Competence, Autonomy, and Relatedness. Marczewski extended this list with the addition of Purpose. Hexad differentiates between six player types: Achiever, Philanthropist, Socializers, Free Spirits, Players, and Disruptors. Later these player types will be discussed in more detail.

Although these typologies were developed using different approaches and for different contexts, they still show clear overlap in the core motivations they describe (Kirchner-Krath et al., 2024). Many of the models include similar motivational themes, such as achievement, exploration, immersion, and social interaction, even if they name or define the types differently. Figure 1 offers a comparative overview of the discussed typologies, mapping each player type to one overarching motivational category. This helps to highlight both where the models differ and where they align, giving a clearer picture of how player types are interpreted across game design and gamification research. The overview is based on the theoretical summary presented in the work of Kirchner-Krath et al. (2024).

	Bartle	**DGD**	**Yee**	**Xu**	**BrainHex**	**Hexad**
Achievement	Achiever	Manager	Achievement	Achiever	Achiever	Achiever
Gaming Intensity & Skill					Survivors Dare Devil	
Exploration & Immersion	Explorer	Wanderer	Immersion Escapism	Free Loader	Mastermind Seeker	Free Spirit
Domination	Killer	Conqueror	Manipulation		Conqueror	Disruptor
Social Interaction	Socializer	Participant	Relationship	Active Buddy Team Player Social Experience Seeker	Socializer	Socializer Philanthropist
External Reward						Player

Figure 1 Theoretical overview of different player typologies, based on Kirchner-Krath et al. (2024).

2.2.2 Hexad typology

As described above, the Hexad typology, described by Marczewski (2015), defines six different player types.

- **Achievers** are motivated by intrinsic motivation of competence. They are the player type that are mostly motivated by mastery and strive to improve themselves**.**
- **Philanthropists,** on the other hand, are driven by altruism and purpose. Their engagement increases when they can help others or contribute to a greater cause, where they experience meaning.
- **Socializers** prioritize social interaction and relatedness. They enjoy gamified systems that encourage collaboration and communication, such as team-based activities.
- **Free Spirits** are autonomy-driven and are motivated by creativity and exploration. They prefer environments that allow for self-expression and open-ended gameplay where they are not held back by rules.
- **Players** are primarily motivated by extrinsic rewards. They engage with gamified systems when they have the opportunity to earn incentives.
- **Disruptors** are mainly driven by change. They tend to test the limits of the system and engage with mechanics that allow them to challenge rules, introduce new ideas, or push boundaries.

Multiple studies related Hexad players to game element preferences. First of all, Tondello et al. (2016) evaluated the game element preferences proposed by Marczewski (2015) in an online survey study. Later, Krath & von Korflesch (2021) tested the same questionnaire as Tondello et al. (2016) on a larger and more diverse sample to verify the results. These four studies found similar results: *Achievers* tend to report enjoying elements that reflect progress and mastery, such as leaderboards, levels, progress bars, and challenges. *Philanthropists* are driven by opportunities to support others, showing a preference for features like gifting, knowledge sharing, organizational roles, and teams. *Socializers* are primarily drawn to features that promote interaction, such as teams, social networks, and competitive or collaborative social activities. *Free Spirits*, on the other hand, are most engaged by creative freedom and exploration, favoring customization, open-ended tasks, and nonlinear gameplay, and like learning and challenge-driven elements. *Players* are most focused on extrinsic rewards such as points, badges, leaderboards, and prizes, but also enjoy levels, progress bars, challenges, certificates, social comparison, social status, and social competition elements. *Disruptors* prefer anarchic gameplay and autonomy, but Krath & von Korflesch (2021) were not able to find a significant effect for innovation platforms and social competition elements. This might be because disruptors are underrepresented in the sample, as this is the least common Hexad player type (Şenocak et al., 2021).

The insights gained from studies on game element preferences have been used in applied studies to test personalized gameful environments tailored to individual Hexad user types. Hallifax et al. (2020) examined the effects of tailored gamification on 258 students in a mathematics learning environment. They compared non-tailored gamification with three tailored approaches: Hexad player profile, their motivational profile based on the pre-test, and a combination of both. The researchers incorporated six gamification elements, i.e., avatars, badges, progress, leaderboards, points, and timers, in an online learning platform and identified the best tailoring approach. Tailoring based on the Hexad profile increased engagement but did not significantly impact motivation, while tailoring based on motivational profile significantly improved intrinsic motivation and helped sustain knowledge-seeking behavior. The most effective approach combined both methods, leading to higher intrinsic motivation and reduced amotivation compared to other groups.

Thompson et al. (2023) focused on using Hexad types to motivate movement in a VR environment. Based on the player's dominant player type, the game environment included a different type of movement and game elements. While the researchers did not find statistically significant results, players in the personalized VR environment indicated higher levels of engagement and motivation, and performed better.

While these player typologies offer valuable insights into how different users engage with gamified systems, understanding their underlying psychological mechanisms requires a more theoretical foundation. The next section builds on this by linking the Hexad framework to different psychological theories. It also explains how these theoretical perspectives inform the selection of relevant PXI dimensions for each player type, providing a structured basis for analyzing player experience and enjoyment.

2.3 Hexad and Psychological Theories

To understand how different users engage with gamified systems, it is important to examine the psychological theories that explain underlying motivational processes. The Hexad player types differ in the motivational needs that drive them (Tondello et al., 2016). By linking Hexad player types to established psychological theories, such as Self-Determination Theory, Uses and Gratifications Theory, Motive Disposition Theory, and the Four Drives Theory, we can gain a deeper understanding of how specific player types experience and respond to gamified environments. These theoretical connections also help clarify why certain PXI dimensions are more relevant to some player types than others.

2.3.1 Self-Determination Theory

Self-Determination Theory (SDT) explains human motivation based on three basic psychological needs, namely autonomy, competence, and relatedness, different people differ in their priority of these needs (Deci & Ryan, 1985). Autonomy is described as the need to feel a sense of ownership and control over one's choices and behaviors. Competence refers to the need to feel effective and capable when facing challenges. Relatedness reflects the need to feel connected to others and to experience a sense of belonging. When these needs are supported, people are more likely to engage in activities out of genuine interest and enjoyment, rather than external pressure or rewards. SDT differentiates between intrinsic motivation, extrinsic motivation, and amotivation (Deci & Ryan, 1985). Intrinsic motivation refers to engaging in an activity for its own sake because it is inherently interesting, enjoyable, or satisfying. Extrinsic motivation, on the other hand, involves performing an activity to achieve a separable outcome, such as receiving a reward, avoiding punishment, or gaining approval. Amotivation refers to a state where individuals do not see a connection between their actions and outcomes, leading to disengagement.

In gamification, SDT is often used to explain why some game elements are more motivating than others (Tyack & Mekler, 2024). For example, Chang (2013) stated that when players are immersed in a gameful environment, they are more intrinsically motivated and more likely to support long-term engagement. Other research also emphasizes the importance of the fulfillment of the three psychological needs (competence, autonomy, and relatedness), on intrinsic motivation (Luarn et al., 2023). However, when these needs are not met or are actively frustrated, gamification can cause need frustration and reduce engagement. Hanus & Fox (2015), for example, argue that gamified systems can shift users' intrinsic motivation to extrinsic learning motivation, potentially undermining deeper engagement.

SDT can be used to understand how different Hexad player types engage with a gamified system. Each Hexad type reflects a different configuration of the psychological needs described by SDT. For example, Achievers are strongly driven by competence and are motivated by progress, goals, and challenge, which aligns with PXI dimensions, such as *mastery, challenge, progress feedback,* and *clarity of goals and rules*. Philanthropists, on the other hand, are primarily motivated by relatedness and purpose. They value social connection and meaningful contributions, which is reflected in their preference for the PXI dimensions *meaning*. By mapping SDT needs onto the motivational focus of each Hexad type, and linking these to relevant PXI dimensions, we gain a clearer understanding of how different players experience and derive enjoyment from gamified systems.

2.3.2 Uses and Gratifications Theory

The Uses and Gratifications Theory (UGT) is a psychological communication theory that focuses on why and how individuals actively choose one media over the others to gratify and meet their needs (Katz et al., 1973). These gratifications can range from entertainment and escapism to social interaction, information seeking, or personal identity (Ruggiero, 2000). Gupta et al. (2024) proposed a new framework that explains why users engage with digital media and games based on six motivations: Information-seeking, Entertainment, Achievement, Novelty-seeking, Social Interaction, and Competition. In the context of gamification, UGT helps explain why different player types are drawn to different experiences. Achievers, for instance, are mainly drawn to achievement and competition. Achievement supports goal-setting, progress tracking, and rewards to motivate and engage players in reaching milestones within a game (Yee et al., 2012). Competition refers to the integration of competitive elements within a gamified system, that is designed to evoke a sense of challenge, comparison, and accomplishment among users (Jaskari & Syrjälä, 2023). In the context of UGT, philanthropists are primarily motivated by social interaction and information-seeking. They are therefore interested in clear goals, well-defined rules, and progress feedback that provides information about their actions (Arya et al., 2024; Gupta et al., 2024). Social interaction is supported through elements that enhance social interactions and help to create and manage social relationships (Arya et al., 2024). This helps philanthropists to feel a deeper meaning.

2.3.3 Motive Disposition Theory

The Motive Disposition Theory (MDT) is another framework to understand how players experience gamified systems (Mcclelland, 1985). MDT assumes that individuals differ in their dispositional needs for achievement, affiliation, autonomy, and power, which shape their preferences, behaviors, and emotional responses (Poeller et al., 2018). The *affiliation* motive reflects a person's need to feel connected to others and avoid social exclusion. The *achievement* motive centers on striving for success and improvement in performance. The *power* motive describes a desire to stand out, lead, or be recognized by others. More recently, *autonomy* has also been recognized as an important motive. Autonomy refers to a person's drive to maintain control over their actions, protect personal boundaries, and grow independently. Additionally, MDT differentiates between implicit and explicit motives (Mcclelland, 1985). Implicit motives can be described as the unconscious motives that the player is not aware of, and are sometimes described as needs (Mcclelland et al., 1989). Explicit motives are the conscious motives that players are aware of, and therefore can be assessed through self-reported questionnaires (Mcclelland et al., 1989). Players are consciously aware of their explicit motives for playing and engaging in a gamified environment. Lastly, motives can be scribed through their approach or avoidance orientation (Elliot, 2013). For example, within the achievement motive, players can be motivated by the hope for success (approach) which might be shown in the enjoyment of challenge, or the fear of failure (avoidance) where players are focused on not performing badly (Bartels & Magun-Jackson, 2009).

A study by Poeller et al. (2018) investigated how implicit and explicit motives relate to player experience in a social game context. They ran an experiment with 109 players and investigated how personality, player motivation, and play style relate to the player's motives in general, after which they focused specifically on the affiliation motive. The findings show that several explicit motives correlate with personality and player motivation, but implicit motives rarely correlate with the explicit measurement in the experiment. Focusing on the implicit affiliation motive, they found that it significantly predicts social behavior in games. Notably, this motive remained a strong predictor even when controlling for player motivation, personality traits, and play style. These results can be used to predict and explain in-game behavior and choices of players.

The underlying motivations of the Hexad player types closely align with the Motive Disposition Theory. For instance, Achievers strongly relate to the achievement motive and therefore value PXI dimensions like *mastery* and *challenge*. Philanthropists, who are more socially oriented, connect with the affiliation motive as they are motivated by helping other players. These players are more likely to experience enjoyment through *meaning*. MDT helps explain how these motivational tendencies shape the way player types engage with and experience gamified environments.

2.3.4 Four Drives Theory

The Four Drives Theory offers another useful perspective for understanding user motivation by identifying four fundamental human drives that guide behavior (Lawrence et al., 2002). These are the drive to acquire, bond, learn, and defend. The *drive to acquire* reflects the desire to obtain a certain resource, status, or reward, and is often associated with competition and achievement. The *drive to bond* captures the need to build relationships and feel a sense of belonging or connection with others. The *drive to learn* refers to the motivation to explore, make sense of the world, and develop new skills or knowledge. Lastly, the *drive to defend* relates to protecting oneself, their values and beliefs, and maintaining a sense of safety and fairness.

This framework can be applied to understand how different Hexad player types interact with gamified systems. Achievers are primarily influenced by the drive to acquire. This drive aligns well with PXI dimensions *mastery* and *challenge*, as these players want to achieve mastery by acquiring the best results, and they like to be challenged by a system. Additionally, Achievers are influenced by the drive to learn, which relates to the PXI dimensions *clarity of goals and rules,* and *progress feedback*. They need to understand the goals and rules of the system to be able to achieve the best results, and they use feedback to keep improving themselves.

Philanthropists, on the other hand, are strongly guided by the drive to bond, focusing on forming connections and contributing to others. This is reflected in their preference for the PXI dimension *meaning*, which supports their desire to make a positive impact within the system. To be able to achieve this they are guided by the drive to learn. They are interested in the *clarity of goals and rules* and *progress feedback* of the system to be able to understand how to contribute and help others. By linking the fundamental human drives to the motivational structure of Hexad types and the experiential components measured by PXI, we gain deeper insight into how gamified systems can be tailored to better meet the needs and expectations of different users.

This section outlined the theoretical foundations linking Hexad player types to underlying psychological needs and motivational drivers. To translate these theoretical insights into measurable constructs, the Player Experience Inventory (PXI) framework provides a practical tool. The next section introduces the PXI in more detail and describes how its dimensions are used to assess player experience.

2.4 Player Experience Inventory

To understand how players engage with gamified systems, it is important to move beyond simply identifying player types and consider how individuals experience games. The Player Experience Inventory (PXI), developed by Vanden Abeele et al. (2020), offers a structured framework to measure player experience across two main dimensions: functional and psychological. The functional dimensions capture a player's direct interaction with a game, which includes the elements *ease of use, progress feedback, audiovisual appeal, challenge,* and *clarity of goals and rules*. The psychological dimension reflects deeper, internal responses and includes *mastery, curiosity, immersion, autonomy,* and *meaning*. Together, these dimensions allow researchers to understand how psychological experiences of players are affected by their interpretation of game design decisions (Vanden Abeele et al., 2020).

The study by Ip & Sweetser (2021) used the PXI to investigate player experiences in virtual reality (VR) games through two stages. In the first stage, they investigated player experiences in VR games through an online survey with 100 participants. They found that Audio-Visual Appeal, Immersion, and Ease of Control are the most important contributors to the VR gaming experience. Interestingly, no relation between VR experience, VR headset, and age was found. In the second part of the study, a remote experiment was conducted with 10 participants to compare player experience in VR and non-VR gameplay. The findings showed that differences between the two formats were mainly explained by the PXI dimensions *immersion, curiosity*, and *progress feedback*. Overall, the results indicated that VR gameplay was experienced more positively than non-VR gameplay. However, the discrepancy between PXI scores and participant preferences suggests that the PXI may not fully capture the unique qualities of VR experiences.

Later, Drey et al. (2022) used the *audiovisual appeal*, *clarity of goals and rules,* and *ease of control* subscales of the PXI model to investigate whether Virtual Reality (VR) can be used for pair-learning. The researchers compared a symmetric and an asymmetric VR pair-learning system, where either both players used a VR headset, or one player a tablet and the other a VR headset. Additionally, they differentiated between a signaling and non-signaling condition, in which important information was highlighted. It was found that the symmetric VR conditions showed better results in terms of presence, immersion, player experience, motivation, fun, and communication. Signaling showed positive results for presence, player experience, and cognitive load. Based on these results the researchers suggested six guidelines for VR pair-learning systems.

This chapter reviewed key literature on gamification, player typologies, psychological theories, and player experience, with a specific focus on the Hexad player type model and the Player Experience Inventory. It demonstrated how different player types align with distinct motivational needs and how these can be linked to specific dimensions of player experience. While prior research has explored tailoring based on player types, no studies have examined yet how specific psychological experiences explain the relationship between player type and enjoyment. Addressing this gap, the current study aims to investigate how Achievers and Philanthropists relate to player enjoyment, and how this relationship is shaped by relevant PXI dimensions. The next chapter describes the methodological approach used to explore these relationships.

3. Method

In my thesis, I investigated how Hexad player type dimensions (specifically Achievers and Philanthropists) relate to player enjoyment, and how this relationship is shaped by specific player experience dimensions measured via the Player Experience Inventory (PXI). To answer this question, I used a large, pre-existing dataset (N = 1,094) collected using the cognitive assessment game "Tunnel Runner" (Markovitch et al., 2024). The data was collected to examine cognitive performance in game-based environments and includes detailed questionnaire responses on player types, player experience, and background variables. The Tunnel Runner dataset is particularly useful to answer my research questions, as it includes both Hexad player type data and PXI player experience ratings, allowing for a direct analysis of the relationships of interest. Additionally, the large sample size of the dataset allows for robust statistical analysis and generalization of findings.

I focused on a subset of PXI dimensions for each player type. As described above, for Achievers I selected the PXI dimensions *mastery, challenge, clarity of goals and rules,* and *progress feedback*. For Philanthropists, *meaning, progress feedback,* and *clarity of goals and rules* were selected as the most relevant dimensions. To investigate how these player experience dimensions shape the relationship between player type and enjoyment, I applied a series of analyses, which will be explained in more detail later in this thesis. By comparing and integrating the findings from different statistical models, the goal was to identify the most accurate and theoretically grounded model of how PXI dimensions together influence the relationship between player type and enjoyment.

3.1 Game Description: Tunnel Runner

Tunnel Runner (Markovitch et al., 2024) is a gamified cognitive assessment game designed to measure executive functioning, such as interference control, response inhibition, and response-rule switching. In the Tunnel Runner, players control a group of rats racing through a circular tunnel filled with obstacles (Figure 2). The main task is to rotate the rats so that the central rat matches the correct color section of the upcoming obstacle. This is done using the A (left) and L (right) keys. Rotation only occurs while a direction key is held down, and the rats keep moving forward automatically, regardless of player input. The perspective of the game moves along with the rats. Good performance is rewarded with points and players receive visual feedback based on the number of successful trials in a row, visualized as the player's streak number. Players must respond quickly and accurately while adapting to different rule sets that are introduced in five different trial types:

- **Baseline trials**. In the baseline trial, all five rats have the same color and the goal of this trial is to measure reaction time and basic performance. Points are given when the central rat moves to the corresponding color on the color wheel. The goal of this trial is to measure participants' reaction time.
- **Mismatching flanker trials**. In this trial, the flanker rats have the opposite color from the central rat on the color wheel. This trial is introduced to assess interference control (Diamond, 2013) by introducing distracting cues. The player receives points when the central rat moves through its corresponding color on the color wheel.
- **Lava trials**. Lava covers the tunnel walls around the rats after a delay. Players have to move the rats to the unaffected area. Players are rewarded with points if the rats do not touch the lava. If the lava is touched, the player will continue to lose points until the rats are moved back to the unaffected area. The goal of this trial is to test response inhibition (Diamond, 2013) by requiring players to stop moving when a stop signal (lava) appears.

- **Ice trials**. During the ice trials, the full tunnel surface is covered with ice, which reverses the game's controls. Pressing A rotates the tunnel to the right, and pressing L rotates it to the left. This trial examines response-rule switching (Schroder et al., 2012) of players as they have to adapt to the control changes.
- **Snacking trials**. Snacking trials are shown during breaks. Players had to choose one of four boxes by pressing a key. Each box could contain either melons, which reward 5 or 10 points, or traps, which take away 5, 25, or 125 points. These trials were included to avoid learning effects on players and to measure decision-making under uncertainty.

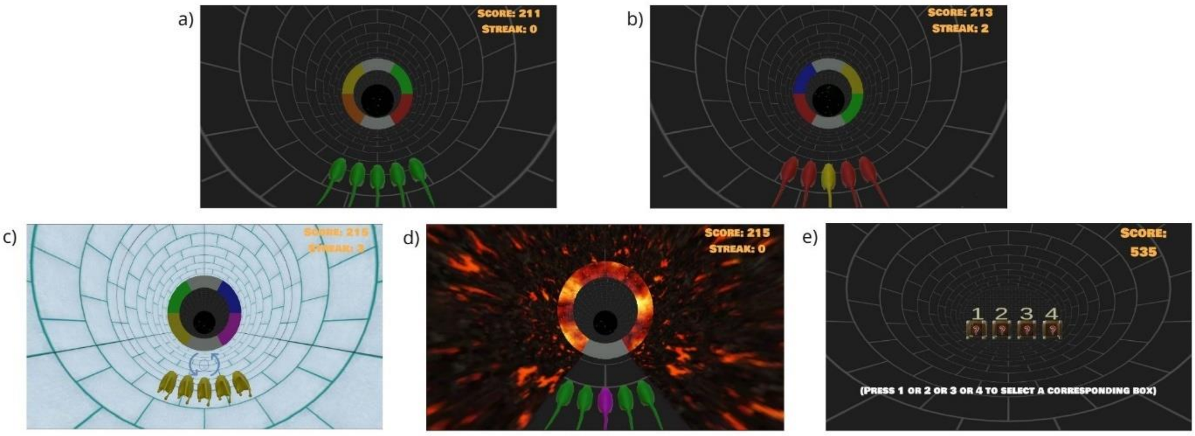

Figure 2: Tunnel Runner trial types. Baseline trial (a), mismatching flanker trial (b), ice trial (c), lava trials (d), and snacking trial (e).

3.2 Participants

The dataset includes data from 1,094 participants recruited through the Connect platform (Hartman et al., 2023). This is a recruitment platform that ensures high data quality using participant screening processes, including demographic verification and attention checks to ensure eligibility and engagement. Participants who failed attention checks or showed no variability in responses were excluded. Of the included sample, 544 identified as men, 522 as women, and 28 as non-binary or self-described. Based on self-reported gaming frequency, 745 were categorized as frequent gamers and 349 as infrequent gamers. Frequent gamers were defined as those who play weekly, while infrequent gamers were those who reported playing less than once a week.

3.3 Materials

For my study, I focused only on the collection of Hexad player types using the Hexad Player Type Questionnaire (Krath et al., 2023), and player experience data collected using the Player Experience Inventory (PXI; Vanden Abeele et al., 2020). The full study, however, included additional measurements of participants' engagement, motivation, and cognitive performance. Engagement was tested through the User Engagement Scale Short Form (UESSF; O'Brien et al., 2018). Participants' perceived effort and frustration were measured through the NASA task load index (TLX; (Hart, 2006). Cognitive abilities were evaluated based on player performance in Tunnel Runner.

3.3.1 Hexad Player Types Questionnaire

Player types were assessed using the Hexad-12 User Types Questionnaire (Krath et al., 2023). This questionnaire is a short version of the Hexad-24 questionnaire developed by Marczewski (2015), and has been validated by Krath et al. (2023). Filling in the questionnaire, participants responded to 12 items on a 7-point Likert scale ranging from 1 ("strongly disagree") to 7 ("strongly agree"). Each Hexad player type is determined using two questionnaire items. The overall player type score is calculated by averaging the responses to those two items. Higher scores indicate a stronger alignment with the motivational orientation described by that player type.

While the Hexad questionnaire measures six player types, this study focused specifically on Achievers and Philanthropists. As mentioned before, these two player types have been selected because of their prominence in the Tunnel Runner dataset. Table 1 shows the distribution of the dominant player types in the dataset. It can be seen that Achievers and Philanthropists are very well represented. It was decided to not include Free Spirits in the focus of this study because it was assumed that these player types may generally experience less enjoyment from the game. Free Spirits are primarily driven by autonomy and tend to enjoy gamified systems that offer a high degree of freedom. However, Tunnel Runner is not designed with autonomy in mind, as the rules of the game are very strict and players lose points if they do not adhere to the rules.

Table 1: Distribution of Player Types in the dataset.

Type	Count	Percentage
Achiever	380	34.73
Free Spirit	251	22.94
Philanthropist	240	21.94
Player	191	17.46
Socializer	18	1.64
Disruptor	14	1.28
Total	1094	

Achievers are motivated by competence and are drawn to gamification elements such as points, badges, and challenges. Tunnel Runner includes elements like points, streaks, and varying difficulty levels, which closely align with Achiever motivations. Philanthropists, on the other hand, are primarily driven by meaning and relatedness. The presence of five rats running together through the tunnel might create a sense of connectedness, as the player is not alone but moves together with the group. Focusing on Achievers and Philanthropists only, allowed for a more in-depth exploration of the relationships, while maintaining a manageable scope for statistical analysis.

The internal consistency (Cronbach's alpha) of the Achiever and Philanthropist subscales were α = .63 and α = .79, respectively. Although the reliability of the Achiever scale is below the commonly accepted threshold of 0.70, the scale has been previously validated in multiple studies (Krath et al., 2023; Şenocak et al., 2021; Tondello et al., 2016). For this reason, I still used the average score of the Achiever scale in the analysis. Appendix A shows an overview of all the items in the Hexad-12 questionnaire.

This thesis treated Hexad types as continuous scales to get a more nuanced understanding of player type tendencies and preferences. The current literature often treats Hexad types as categorical, assigning participants to their dominant type. However, as previous research has shown, player types tend to overlap and are often correlated (Kirchner-Krath et al., 2024). Tondello et al. (2019) also emphasized that player typologies place individuals into fixed categories, while most players show a mix of motivations rather than just one dominant type.

3.3.2 Player Experience Inventory (PXI)

Player experience was tested through 30 questions from the Player Experience Inventory (PXI; Vanden Abeele et al., 2020). The PXI measures how players have experienced a game or gamified system through 10 dimensions: *mastery, curiosity, immersion, autonomy, meaning, ease of control, progress feedback, audiovisual appeal, challenge,* and *clarity of goals and rules*, divided into functional and psychosocial levels. The functional level includes *ease of control, progress feedback, audiovisual appeal, challenge,* and *clarity of goals and rules*. The psychosocial level includes *mastery, curiosity, immersion, autonomy,* and *meaning*. Enjoyment is the outcome variable representing player experience in the model. The *audiovisual appeal* items were excluded from the survey as they overlap with the aesthetic appeal measurement in the UESSF scale (O'Brien et al., 2018). The PXI measurements are all Likert scale items ranging from 1 (strongly disagree) to 7 (strongly agree), with 4 being neutral. Each dimension was assessed using three items, and the average of these items was used as the overall dimension score in the analysis. In Appendix B you will find an overview with the specific questions for each item.

Achievers are primarily motivated by competence, have a strong drive to perform well and achieve optimal in-game results, which is reflected in the psychosocial level *mastery*. This motivation leads them to seek out challenges, clear goals and structured rules that define how the system works, and they value receiving clear feedback on their progress, which helps them develop a sense of mastery. In order to achieve mastery, Achievers are therefore mostly influenced by the functional dimensions *clarity of goals and rules, progress feedback,* and *challenge*. As a result, their overall experience is mostly shaped by the PXI dimensions of *mastery, challenge, clarity of goals and rules,* and *progress feedback*. Philanthropists, on the other hand, are mainly driven by relatedness and are motivated by a sense of purpose and social connection. This is reflected in the psychosocial dimension *meaning* and the functional dimensions *progress feedback,* and *clarity of goals and rules*. For both types, *clarity of goals and rules,* and *progress feedback* are important for understanding how to navigate the system, either to master tasks or help others, and receiving guidance on their performance. Feedback plays a crucial role in player experience because it helps individuals understand how well they are doing in relation to the goals and rules of the game. For Achievers, feedback supports their need for competence by providing clear indicators of progress and performance. For Philanthropists, feedback can also offer reassurance that their actions are meaningful or aligned with the system's purpose, even if their motivation is less performance-driven.

The internal consistency of the included PXI subscales was tested using Cronbach's alpha. The results indicated acceptable to excellent reliability: mastery (α = .88), meaning (α = .92), challenge (α = .84), progress feedback (α = .77), clarity of goals and rules (α = .82), and enjoyment (α = .96). These values suggest that the subscales are reliable measures of their respective player experience dimensions.

3.4 Data Analysis

This section describes the data analysis process used to investigate how PXI dimensions help explain the relationship between Achiever and Philanthropist scores, and enjoyment. It begins with assumption checking, followed by an explanation of the analytical approach used to examine how PXI dimensions relate to enjoyment for Achievers and Philanthropists. Finally, the criteria used to evaluate model fit are described.

3.4.1 Assumption Checks

All analyses were conducted using JASP. Before the main analyses, statistical assumptions were checked including normality, linearity, multicollinearity, homoscedasticity, and outliers. Normality was assessed using the Shapiro-Swilk test and visual inspection of Q-Q plots. Linearity and homoscedasticity were evaluated by plotting residuals against predicted values. Homoscedasticity has additionally been evaluated through residual covariance tables of each SEM model. Multicollinearity was examined using variance inflation factor (VIF) scores and correlations between the variables were assessed to check for no large correlations. As all the variables used in the analyses are measured through Likert scales, there were no specific outliers. Participants can answer to the extremes, but this is not viewed as an outlier.

The assumption checks indicated that the assumptions of normality and linearity were not fully met, which is not uncommon when working with Likert-scale data. Although some variables showed non-normal distributions, no data transformation was applied. Since the data are based on Likert scales, which are ordinal by nature, transforming the data could reduce interpretability. Instead, the analyses were conducted using the Robust Maximum Likelihood (MLR) estimator. This approach provides robust standard errors and fit indices that are less sensitive to violations of normality and linearity (Bryan et al., 2007; Rosseel, 2017). Additionally, bootstrapping with 5,000 samples was applied to obtain more accurate confidence intervals for the indirect effects and to correct for potential deviations from normality.

3.4.2 Analytical Approach

To examine how PXI dimensions relate to enjoyment for Achievers and Philanthropists, I systematically tested and refined mediation, moderation, and moderated mediation models using Structural Equation Modeling (SEM). Initially, simple mediation and moderation analyses were conducted separately to assess whether PXI dimensions mediate or moderate the relationship. If evidence of both mediation and moderation had emerged, moderated mediation models would have been tested. After this, final models were developed for both the Achiever and Philanthropist profiles, and modification indices, which suggest additional relationships between existing variables in the model, were reviewed to identify potential improvements to model fit. During this process, it was ensured that all modifications were theoretically justified. The analysis was inherently iterative: based on the results obtained at each step, subsequent analyses were adjusted accordingly. Below a more detailed description of each of the analyses is given.

Analysis 1: Mediation

I first explored whether the relationship between player type scores (i.e., Achiever or Philanthropist) and enjoyment was mediated by specific PXI dimensions. In mediation analysis, the goal is to determine whether the effect of an independent variable (in this case, player type) on a dependent variable (enjoyment) operates through a third variable, known as the mediator (here, PXI dimensions such as mastery or meaning). If mediation is present, it suggests that the player type influences the individual PXI dimensions, which in turn influences enjoyment. This provides insight into the underlying mechanisms of the relationship. A visual representation of a mediation framework is shown in Figure 3**.**

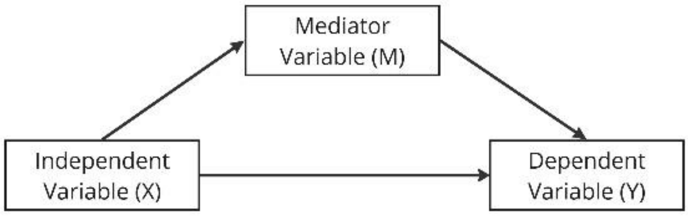

Figure 3: Visual Overview of a Mediation Model

First, simple mediation models were run for each player type separately. Each PXI dimension (*mastery, meaning, challenge, progress feedback*, and *clarity of goals and rules*) was first tested individually to examine if it mediated the relationship between player type and enjoyment.

Analyses 2: Moderation

Next, moderation models were evaluated to test whether PXI dimensions moderated the relationship between player type scores and enjoyment. In moderation analysis, the aim is to determine whether the effect of an independent variable on an outcome variable changes depending on the level of a third variable, called the moderator. In this case, the analysis examined whether the relationship between player type scores (Achiever or Philanthropist) and enjoyment varied depending on each PXI dimension. If moderation is supported, it suggests that the influence of player type on enjoyment is conditional on the player's experience within a specific PXI dimension. A visual representation of a moderation model is shown in Figure 4.

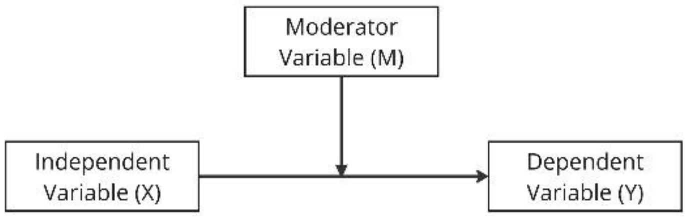

Figure 4: Visual Overview of a Moderation Model

To test this, all variables were mean-centered, and interaction terms between the centered player type scores and PXI dimensions were created. Multiple simple moderation models were conducted, each including one PXI dimension as a moderator, to test whether the relationship between player type scores and enjoyment changed depending on that specific experience dimension.

Analyses 3: Moderated Mediation

This analysis was only performed if both mediation and moderation effects were found. Moderated mediation analysis examines whether the indirect effect of an independent variable (in this case, player type) on an outcome variable (enjoyment) through a mediator depends on the level of another variable (the moderator). Figure 5 shows a visual representation of a moderated mediation model.

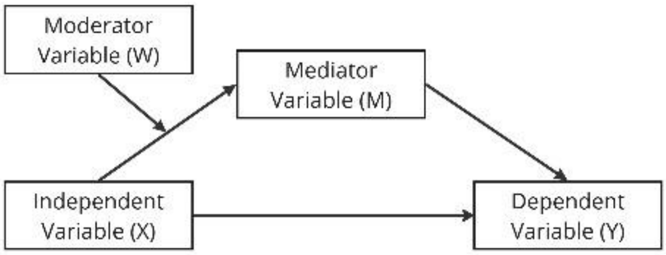

Figure 5: Visual Overview of a Moderated Mediation Model

In this study, we tested whether the indirect effect of player type score on enjoyment through one PXI dimension varied depending on the level of another PXI dimension. In other words, one PXI dimension served as the mediator between player type and enjoyment, while another PXI dimension influenced (moderated) this mediation effect. This made it possible to explore how different combinations of player experiences influence the relationship between player type score and enjoyment.

Analyses 4: Model Development

After conducting the previous analyses, final models with the best fit using several fit criteria, i.e., χ^2, CFI, RMSEA, and SRMR, were iteratively developed for both the Achiever and Philanthropist profiles. The process began by constructing initial models that included all relevant PXI dimensions identified in the earlier analyses. Next, modification indices were examined to explore possible improvements in model fit, focusing only on relationships between included variables. Importantly, only modifications that were theoretically justified were incorporated. This was an iterative process, where model evaluation criteria guided each step. The final models were the ones that showed the best fit for each player type profile.

3.4.3 Model Evaluation Criteria

To evaluate the fit of the models, multiple fit indices were considered. Specifically, model fit was assessed using the chi-squared (χ^2) test, Comparative Fit Index (CFI), Root Mean Square Error of Approximation (RMSEA), and the Standardized Root Mean Square Residual (SRMR). These indices have been recommended before to evaluate SEM models (Bryan et al., 2007; Hu & Bentler, 1999). While a non-significant chi-squared test indicates good model fit, this test is sensitive to sample size (Bryan et al., 2007). Therefore, greater emphasis was placed on the evaluation of CFI, RMSEA, and SRMR. Following commonly accepted thresholds (Hu & Bentler, 1999), CFI values above .95, RMSEA values close to or below .06, and SRMR values below .08 were considered indicative of good model fit. Model modifications were guided by both theoretical reasoning and examination of modification indices when initial fit indices were suboptimal.

4. Results

To explore how differences in player experience account for the relationship between player type scores and enjoyment, multiple theoretical models were tested. This section presents the results of the model development process. It begins with the outcomes of the simple mediation analyses for both the Achiever and Philanthropist models. This is followed by the results of the simple moderation analyses, which examine how PXI dimensions influence the relationship between player type and enjoyment. The last subsection outlines how the final models were constructed, based on the findings from the mediation and moderation analyses, and discusses the reasoning behind the chosen model structures.

4.1 Mediation Analysis

To examine whether PXI dimensions mediate the relationship between player type and enjoyment, separate mediation analyses were conducted for each relevant PXI dimension. Each analysis included the player type score (Achiever or Philanthropist) as the predictor (X), the PXI dimension as the mediator (M), and enjoyment as the outcome (Y).

Achiever Mediation Analyses

Four simple mediation models were conducted with the Achiever score as the predictor and each of the following PXI dimensions as mediators: *challenge, clarity of goals and rules, progress feedback*, and *mastery*. Figure 6 shows the path diagrams with standardized path estimates and significance levels. Table 2 presents the corresponding indirect effect estimates for each model with the 95% bootstrapped confidence intervals.

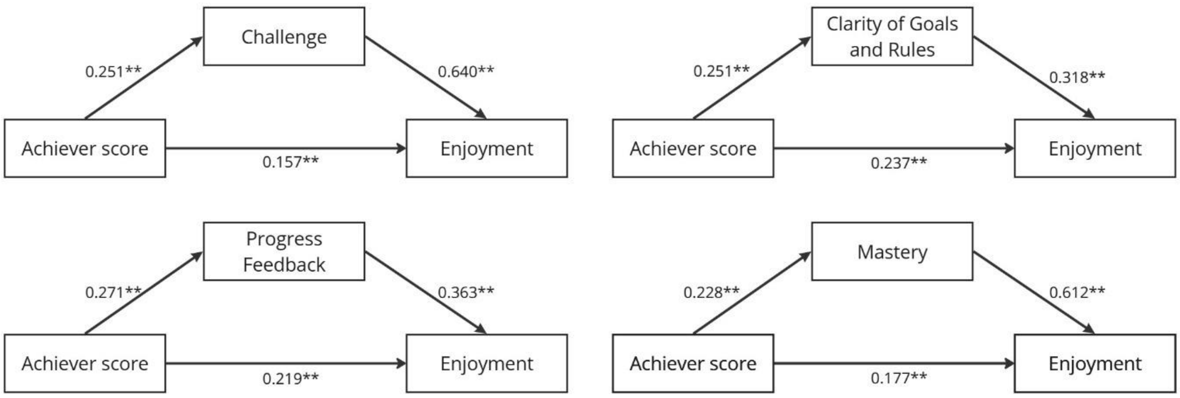

*Figure 6: Path diagrams of individual mediation analysis for Achiever scores including path estimates, *p < .05; **p < .01*

Table 2: Standardized estimates of indirect effects of the simple mediation models for Achievers

PXI dimension	Indirect effect	p-value	95% confidence interval	
			Lower	Upper
Challenge	0.160	< .001	0.117	0.203
Clarity of Goals and Rules	0.080	< .001	0.054	0.106
Progress Feedback	0.098	< .001	0.067	0.129
Meaning	0.140	< .001	0.100	0.179

As shown in Figure 6, Achiever scores significantly predicted each of the PXI dimensions i.e., *challenge, clarity of goals and rules, progress feedback,* and *mastery*. In turn, each PXI dimension significantly predicted enjoyment, controlling for Achiever score. Table 2 summarizes the indirect effects for all models, along with the corresponding 95% bootstrapped confidence intervals. All four indirect effects were statistically significant, indicating that *challenge, clarity of goals and rules, progress feedback*, and *mastery* each mediated the relationship between Achiever score and enjoyment. The direct effect of Achiever score on enjoyment remained significant across all models, with a consistent total effect of 0.317 ($p < .001$), suggesting partial mediation in all cases.

Philanthropist Mediation Analyses

Three simple mediation models were conducted with Philanthropist score as the predictor and the following PXI dimensions as mediators: *clarity of goals and rules, progress feedback*, and *meaning*. Figure 7 displays the path diagrams with corresponding standardized path estimates and significance levels. Table 3 presents the corresponding indirect effect estimates for each model with the 95% bootstrapped confidence intervals.

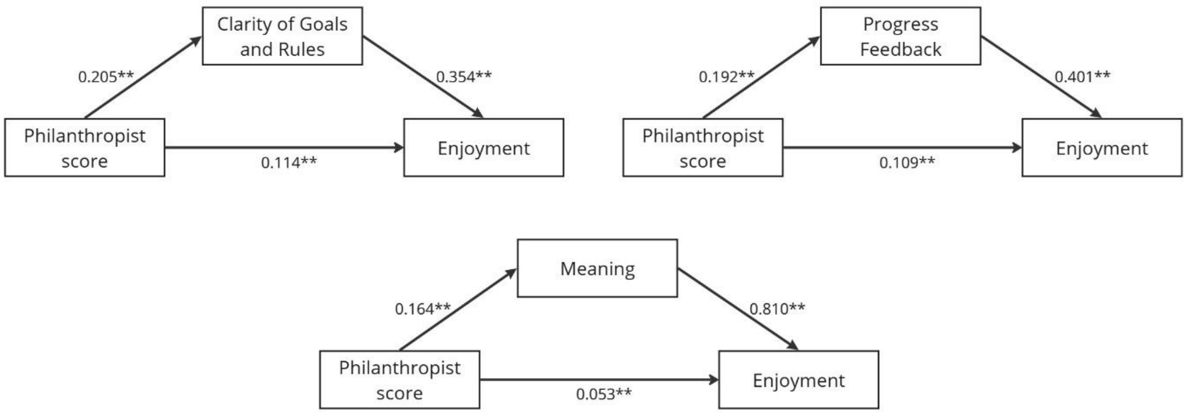

*Figure 7: Path diagrams of individual mediation analysis for Philanthropist scores, *p < .05, **p < .001*

Table 3: Standardized estimates of indirect effects of the simple mediation models for Philanthropists

PXI dimension	**Indirect effect**	**p-value**	**95% confidence interval**	
			Lower	Upper
Clarity of Goals and Rules	0.072	< .001	0.044	0.101
Progress Feedback	0.077	< .001	0.048	0.106
Meaning	0.133	< .001	0.083	0.184

As shown in Figure 7, Philanthropist scores significantly predicted all three PXI dimensions i.e., *clarity of goals and rules, progress feedback,* and *meaning*. Each of these dimensions, in turn, significantly predicted enjoyment, controlling for the direct effect of Philanthropist score. All three indirect effects were statistically significant, indicating that *clarity of goals and rules, progress feedback*, and *meaning* each mediate the relationship between Philanthropist score and enjoyment. The direct effect of Philanthropist score on enjoyment remained significant in all models, with a consistent total effect of 0.186 ($p < .001$). These findings support partial mediation for all three PXI dimensions.

4.2 Moderation Analysis

To examine whether the relationship between either Achiever or Philanthropist player type scores and enjoyment was moderated by specific PXI dimensions, a series of moderation analyses were conducted. For each analysis, all predictor variables were mean-centered to reduce multicollinearity. Interaction terms were then computed by multiplying the centered player type score with the relevant centered PXI dimension. Each model included three predictors: the centered player type score (X), the centered PXI dimension (M), and their interaction term (X × M), with enjoyment as the dependent variable (Y). The significance of the interaction term indicates the presence of a moderation effect.

Achiever Moderation Analyses

Four simple moderation models were conducted with Achiever score as the predictor and each of the following PXI dimensions as moderators: *challenge, clarity of goals and rules, progress feedback*, and *mastery*. In each model, interaction terms were created using centered variables.

Table 4 presents the standardized coefficients for the interaction terms, along with their 95% bootstrapped confidence intervals and p-values. None of the interaction effects were statistically significant. Additionally, all interaction terms had narrow bootstrapped confidence intervals. This suggests that if any interaction between Achiever score and the individual PXI dimensions exists, it is likely to be small. In other words, the data do not support the presence of moderation effects, indicating that the relationship between Achiever score and enjoyment does not vary across the different PXI dimensions. Based on these results, no multiple moderation or moderated mediation models were estimated for the Achiever profile.

Table 4: Standardized estimates of interaction terms of the simple moderation models for Achievers

PXI dimension	**Interaction Term**	**p-value**	**95% confidence interval**	
			Lower	Upper
Challenge	-0.038	0.051	-0.077	2.350×10^{-4}
Clarity of Goals and Rules	0.014	0.588	-0.037	0.065
Progress Feedback	0.028	0.304	-0.025	0.082
Mastery	0.034	0.478	-0.060	0.129

Philanthropist Moderation Analyses

To test whether the relationship between Philanthropist score and enjoyment was moderated by specific PXI dimensions, three separate moderation models were tested. The PXI dimensions included in these analyses were *clarity of goals and rules, progress feedback,* and *meaning*. All variables were mean-centered before computing interaction terms between the Philanthropist score and each PXI dimension. Table 5 shows the standardized coefficients for the interaction terms, along with their 95% bootstrapped confidence intervals and p-values for each model. None of the interaction effects reached statistical significance, and the bootstrapped confidence intervals for all interaction terms were narrow. This indicates that any potential interaction between Philanthropist score and the individual PXI dimensions is likely minimal. Overall, the data did not support moderation effects and therefore the relationship between Philanthropist score and enjoyment does not vary as a function of the individual PXI dimensions. Based on these findings, no further multiple moderation or moderated mediation models were conducted for the Philanthropist profile.

Table 5: Standardized estimates of interaction terms of the simple moderation models for Philanthropists

PXI dimension	**Interaction Term**	**p-value**	**95% confidence interval**	
			Lower	Upper
Clarity of Goals and Rules	0.018	0.523	-0.037	0.073
Progress Feedback	-0.018	0.534	-0.076	0.039
Meaning	-0.007	0.675	-0.042	0.027

4.3 Model Development

So far it has been established that PXI dimensions individually mediate the relation between player type scores and enjoyment for both player type profiles. The moderation analyses did not show significant results. Therefore, a multiple mediation analysis was conducted to develop the final models for both the Achiever and Philanthropist profiles.

Achiever Multiple Mediation

After examining the individual mediation models for each PXI dimension, an initial multiple mediation model was constructed to test a more comprehensive representation of how PXI dimensions mediate the relationship between Achiever scores and enjoyment. The multiple mediation model included all four PXI dimensions relevant to Achievers, *challenge, clarity of goals and rules, progress feedback,* and *mastery*, as parallel mediators. Figure 8 shows the path diagrams with standardized path estimates and significance levels.

Table 6 includes all direct, indirect, and total effects, along with 95% bootstrapped confidence intervals and significance levels. Fit indices are presented in Table 7. The initial model did not include modifications.

The overall multiple mediation model demonstrated poor fit to the data, $\chi^2(6) = 969.559$, $p < .001$, CFI = .451, RMSEA = .383, SRMR = .258 (see Table 7). These values fall well outside commonly accepted thresholds for good model fit, indicating that the proposed model does not adequately capture the underlying structure of the data. As mentioned before, Hu & Bentler (1999) defined good model fit criteria as CFI ≥ .95, RMSEA ≤ .06, and SRMR ≤ .08. Given this poor model fit, the results of the model should be interpreted with caution. For the sake of transparency, the results are still presented below, but they are not used as a basis for the main conclusions of this study.

The multiple mediation model revealed a significant total indirect effect of Achiever score on enjoyment through the PXI dimensions (β = .228, p < .001), indicating that these dimensions collectively mediate the relationship. However, the direct effect of Achiever score on enjoyment remained significant (β = .130, p = < .001), suggesting partial mediation. Interestingly, the direct relation between progress feedback to enjoyment was found to be insignificant, resulting in an insignificant indirect effect for progress feedback. Because of the poor model fit, modifications were explored to improve the model and better represent the structure of player experience dimensions.

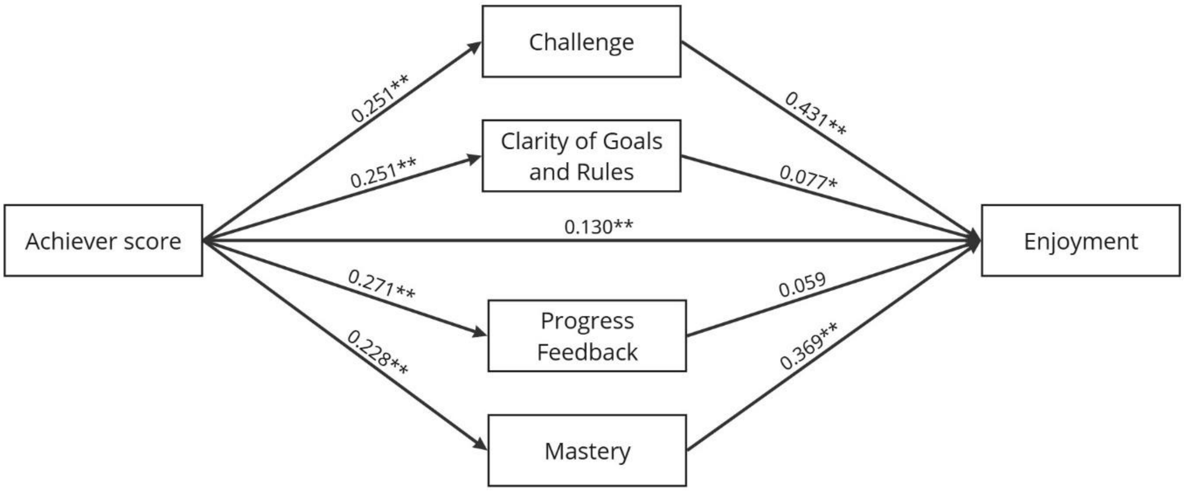

*Figure 8: Path diagram of a multiple mediation model for Achievers, *p < .05, **p < .001*

Table 6: Standardized estimates of direct, indirect, and total effect of multiple mediation model for Achievers

Path	**Std. estimate (β)**	**SE**	**p-value**	**95% Confidence Interval**	
				Lower	Upper
Achiever → Challenge	0.251	0.034	< .001	0.184	0.317
Achiever → Clarity of Goals & Rules	0.251	0.034	< .001	0.184	0.318
Achiever → Progress Feedback	0.271	0.035	< .001	0.202	0.339
Achiever → Mastery	0.228	0.032	< .001	0.166	0.290
Challenge → Enjoyment	0.431	0.038	< .001	0.357	0.505
Clarity of Goals & Rules → Enjoyment	0.077	0.029	0.008	0.020	0.134
Progress Feedback → Enjoyment	0.059	0.036	0.105	-0.012	0.130
Mastery → Enjoyment	0.369	0.037	< .001	0.296	0.442
Indirect Effect Challenge	0.108	0.017	< .001	0.074	0.142
Indirect Effect Clarity of Goals and Rules	0.019	0.008	0.013	0.004	0.035
Indirect Effect Progress Feedback	0.016	0.010	0.110	-0.004	0.036
Indirect Effect Mastery	0.084	0.015	< .001	0.055	0.113
Total Indirect Effect	0.228	0.026	< .001	0.177	0.278
Direct Effect	0.130	0.025	< .001	0.081	0.179
Total Effect	0.357	0.033	< .001	0.292	0.442

Table 7: Model fit indices for multiple mediation model for Achievers

	Baseline test			**Fit Indices**		
	χ^2	**df**	**p-value**	**CFI**	**RMSEA**	**SRMR**
Multiple Mediation Model	969.559	6	<.001	0.451	0.383	0.258

The first model modification involved adding a correlational path between the residuals of *clarity of goals and rules* and *progress feedback*. This accounts for shared variance between the two dimensions that is not explained by other paths in the model. The relationship is theoretically sound, as understanding the goals and rules of a game provides the necessary context for interpreting performance feedback (Weitze, 2014). Without a clear understanding of what the game expects, players may struggle to make sense of the feedback they receive. Conversely, high-quality feedback helps reinforce or clarify game goals by showing players whether they are on the right track, highlighting successful strategies, or pointing out mistakes (Johnson et al., 2017). This creates a mutual relationship between the two dimensions.

A second model modification involved adding a correlational path between the residuals of *challenge* and *mastery*. This allowed the model to account for shared variance between the two dimensions that was not captured by existing paths and also aligns well with existing theory. Experiencing mastery typically requires an appropriate level of challenge. When tasks are too easy, they do not create a sense of accomplishment, while overly difficult tasks can lead to frustration instead of growth (Juul, 2009). When players face a challenge and successfully overcome it, they are more likely to experience a sense of competence and mastery (Denisova et al., 2017). In this way, the two are closely intertwined in player experiences.

These modifications significantly improved the model fit. To further refine the model, additional causal paths were added from *clarity of goals and rules* and *progress feedback* to both *challenge* and *mastery*. This was theoretically supported, as *clarity of goals and rules* and *progress feedback* can be seen as more foundational experiences that players need to be able to engage in a gamified system (Vanden Abeele et al., 2020). These lower-level elements enable players to effectively engage with a gamified system and shape how they experience the more higher-level dimensions like *challenge* and *mastery.* When players understand what they are supposed to do and receive ongoing feedback, they are better able to engage with the challenge of the game (Denisova et al., 2017; Weitze, 2014). Feedback, and goals and rules help define the task and provide structure, which makes the challenge feel more manageable and purposeful (Johnson et al., 2017; Weitze, 2014). These dimensions also play an important role in enabling mastery. In order to feel a sense of competence or growth, players need to understand what they are working toward and how well they are doing (Weitze, 2014). In this way, *clarity of goals and rules* and *progress feedback* act as conditions that allow players to experience *challenge* and *mastery.* These modifications also nicely fit the previously added correlations, suggesting that *clarity of goals and rules* and *progress feedback* formed one layer, while *challenge* and *mastery* formed another layer. The relationship between *progress feedback* to enjoyment remained insignificant with these modifications. Additionally, it showed slightly worse AIC and BIC scores, and therefore it was removed from the model.

The final multiple mediation model including the previously explained modifications is presented in Figure 9, with standardized path estimates and significance levels. All direct, indirect, and total effects are reported in Table 8. Fit indices are presented in Table 9Table 9. The final model demonstrated excellent fit to the data, $\chi^2(1) = 2.604$, $p = .107$, CFI = .999, RMSEA = .038, SRMR = .006 (see Table 9). These values indicate a well-fitting model according to commonly accepted guidelines. All included paths were statistically significant and aligned with the proposed theoretical structure. The model showed a total indirect effect $\beta = .31$, $p < .001$. The direct effect remained significant, $\beta = .119$, $p < .001$, indicating partial mediation. Overall, the final model provides a coherent framework for understanding how players' Achiever score indirectly relates to enjoyment through specific player experience dimensions.

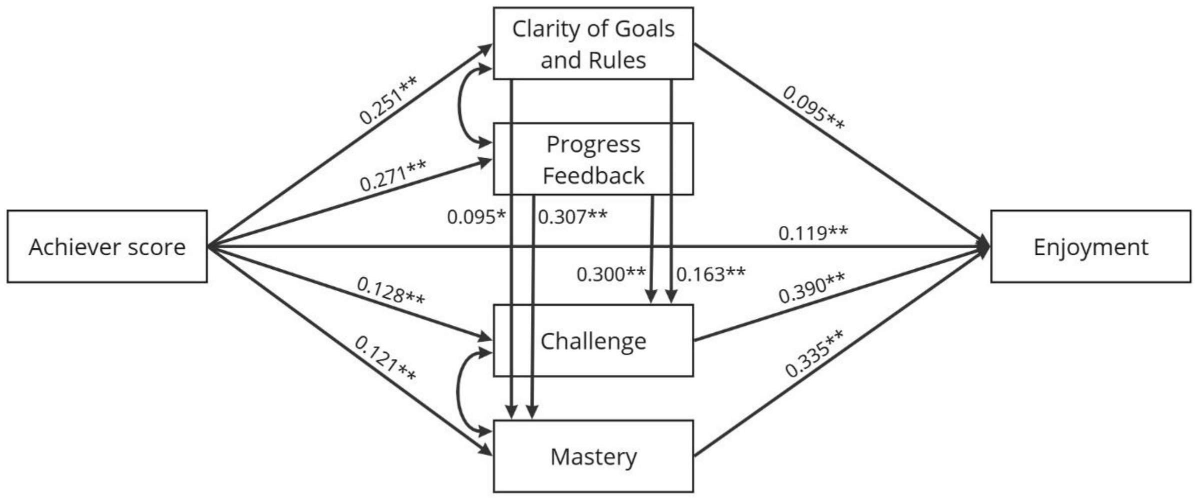

*Figure 9: Path diagram of final Achiever model, *p < .05, **p < .001*

Table 8: Standardized estimates of direct, indirect, and total effects of final Achiever model

Path	Std. estimate (β)	SE	p-value	95% Confidence Interval	
				Lower	Upper
Achiever → Clarity of Goals & Rules	0.251	0.034	<.001	0.184	0.318
Achiever → Progress Feedback	0.271	0.035	<.001	0.202	0.339
Achiever → Challenge	0.128	0.036	<.001	0.059	0.198
Achiever → Mastery	0.121	0.032	<.001	0.059	0.183
Clarity of Goals & Rules → Enjoyment	0.095	0.023	<.001	0.049	0.140
Challenge → Enjoyment	0.390	0.033	< .001	0.325	0.454
Mastery → Enjoyment	0.335	0.032	<.001	0.273	0.397
Clarity of Goals & Rules → Challenge	0.163	0.036	< .001	0.091	0.234
Clarity of Goals & Rules → Mastery	0.095	0.034	0.006	0.027	0.162
Progress Feedback → Challenge	0.300	0.039	< .001	0.223	0.377
Progress Feedback → Mastery	0.307	0.036	< .001	0.237	0.378
Indirect Effect Challenge	0.050	0.015	< .001	0.021	0.079
Indirect Effect Clarity of Goals and Rules	0.024	0.007	< .001	0.011	0.037
Indirect Effect Mastery	0.041	0.011	< .001	0.018	0.063
Indirect Clarity of Goals and Rules through Challenge	0.016	0.004	< .001	0.008	0.024
Indirect Clarity of Goals and Rules through Mastery	0.008	0.003	0.011	0.002	0.014
Indirect Progress Feedback through Challenge	0.027	0.006	< .001	0.015	0.039
Indirect Progress Feedback through Mastery	0.032	0.006	< .001	0.020	0.045
Total Indirect Effect	0.198	0.023	< .001	0.153	0.242
Direct Effect	0.119	0.022	< .001	0.075	0.163
Total Effect	0.317	0.030	< .001	0.258	0.377

Table 9: Model fit indices for final Achiever model

	Baseline test			Fit Indices		
	χ^2	df	p-value	CFI	RMSEA	SRMR
Final Achiever Model	2.604	1	0.107	0.999	0.038	0.006

Philanthropist Multiple Mediation

Following the individual mediation analyses for each PXI dimension, an initial multiple mediation model was developed to provide a more integrated view of how PXI dimensions mediate the relationship between Philanthropist scores and enjoyment. This multiple mediation model included the PXI dimensions *clarity of goals and rules, progress feedback*, and *meaning*, each included as a parallel mediator. Figure 10 shows the path diagrams with standardized path estimates and significance levels. All direct, indirect, and total effects are presented separately in Table 10, along with 95% bootstrapped confidence intervals and significance levels. Fit indices are presented in Table 11. This initial model did not include modifications.

The initial multiple mediation model showed poor model fit, $\chi^2(3) = 549.407$, $p < .001$, CFI = .630, RMSEA = .408, SRMR = .219 (see Table 11). These indices fall outside accepted thresholds for good fit, indicating that the model does not adequately represent the data. Given this, the results of the model should be interpreted with caution. For the sake of transparency, the results are still presented below, but they are not used as a basis for the main conclusions of this study.

The model showed that all indirect effects from Philanthropist score to enjoyment through the PXI dimensions were statistically significant, with the total indirect effect $\beta = .161$, $p < .001$. The direct effect was insignificant, $\beta = .033$, $p = .113$, suggesting total mediation. In order to enhance model fit and more accurately capture the fundamental structure of player experience dimensions, multiple modifications were investigated.

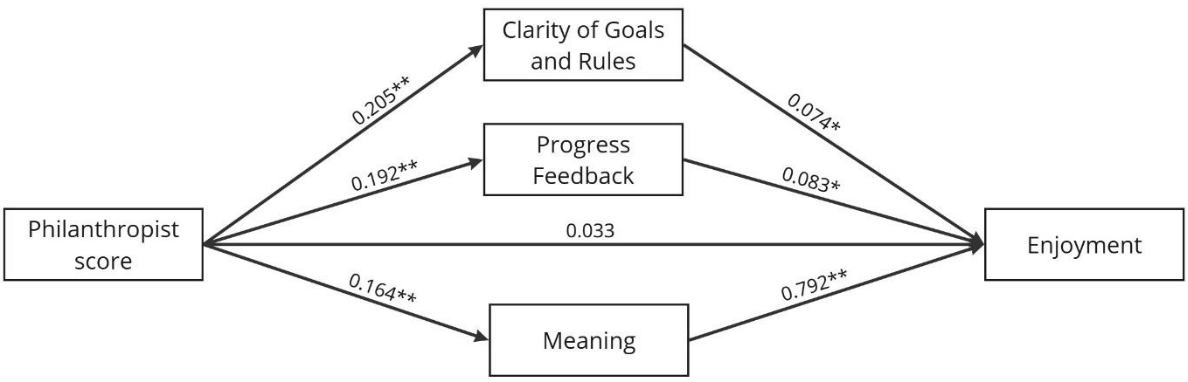

*Figure 10: Path diagram of multiple mediation model for Philanthropists, *p < .05, **p < .001*

Table 10: Standardized estimates of direct, indirect, and total effect of multiple mediation model for Philanthropists

Path	**Std. estimate (β)**	**SE**	**p-value**	**95% Confidence Interval**	
				Lower	Upper
Philanthropist → Clarity of Goals & Rules	0.205	0.036	< .001	0.134	0.275
Philanthropist → Progress Feedback	0.192	0.033	< .001	0.127	0.256
Philanthropist → Meaning	0.164	0.032	< .001	0.102	0.227
Clarity of Goals & Rules → Enjoyment	0.074	0.026	0.005	0.023	0.125
Progress Feedback → Enjoyment	0.083	0.029	0.004	0.026	0.140
Meaning → Enjoyment	0.792	0.014	< .001	0.764	0.820
Indirect Effect Clarity of Goals and Rules	0.015	0.006	0.014	0.003	0.027
Indirect Effect Progress Feedback	0.016	0.006	0.009	0.004	0.028
Indirect Effect Meaning	0.130	0.025	< .001	0.081	0.179
Total Indirect Effect	0.161	0.028	< .001	0.107	0.216
Direct Effect	0.033	0.021	0.113	-0.008	0.074
Total Effect	0.194	0.035	< .001	0.126	0.263

Table 11: Model fit indices for multiple mediation model for Philanthropists

	Baseline test			**Fit Indices**		
	χ^2	df	p-value	CFI	RMSEA	SRMR
Multiple Mediation Model	549.407	3	<.001	0.630	0.408	0.219

The first model modification involved adding a correlational path between the residuals of *clarity of goals and rules* and *progress feedback*. As explained above in the Achiever model, clear goals and rules ensure the understandability of the provided feedback in the system (Weitze, 2014), and progress feedback can enhance a player's understanding of the goals and rules in the system (Johnson et al., 2017). In this way, the two are closely intertwined in player experiences.

Additionally, modification indices suggested causal relationships from *clarity of goals and rules* and *progress feedback* to *meaning*. This is also supported by theory, as *clarity of goals and rules* and *progress feedback* are part of the more lower-level functional dimensions of the PXI model (Vanden Abeele et al., 2020). These are the direct consequences of game design choices and lead to higher-level psychosocial dimensions, such as meaning. Understanding how the system works (e.g. its goals and rules) and how well they are doing (e.g. progress feedback), plays an important part in experiencing a sense of meaning (Treanor et al., 2011).

These modifications were added to the model and resulted in a final model to explain the mediating relation of the PXI dimensions in the relationship between Philanthropist score and enjoyment. Figure 11 shows the path diagrams with standardized path estimates and significance levels. All direct, indirect, and total effects are presented separately in

Table 12, along with 95% bootstrapped confidence intervals and significance levels. Fit indices are reported in Table 13. Fit indices indicated that the model is just-identified (χ^2 = 0.000, *df* = 0, CFI = 1.000). As a result, fit indices such as RMSEA and SRMR were not available. In a just-identified model, the number of estimated parameters is equal to the number of known data points, resulting in mathematically perfect fit indices. As the model structure is relatively simple and all relationships in the model are supported by theory, this model can still be considered acceptable and meaningful within the context of this study. As shown in Table 12, all included paths were statistically significant and aligned with the proposed theoretical structure. The model showed a total indirect effect $\beta = .154$, $p < .001$. The direct effect remained insignificant, $\beta = .032$, $p = .114$, indicating total mediation. Overall, the final model provides a coherent framework for understanding how players' Philanthropist score indirectly relates to enjoyment through specific player experience dimensions.

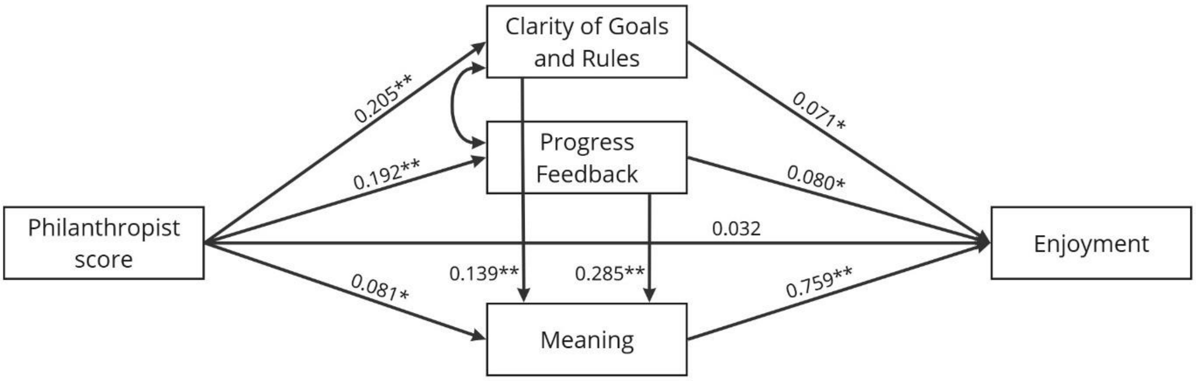

*Figure 11: Path diagram of final Philanthropist model, *p < .05, **p < .001*

Table 12: Standardized estimates of direct, indirect, and total effect of final Philanthropist model

Path	**Std. estimate (β)**	**SE**	**p-value**	**95% Confidence Interval**	
				Lower	Upper
Philanthropist → Clarity of Goals & Rules	0.205	0.036	< .001	0.134	0.275
Philanthropist → Progress Feedback	0.192	0.032	< .001	0.127	0.256
Philanthropist → Meaning	0.081	0.031	0.009	0.020	0.142
Clarity of Goals & Rules → Enjoyment	0.071	0.025	0.004	0.022	0.119
Clarity of Goals & Rules → Meaning	0.139	0.033	< .001	0.074	0.205
Progress Feedback → Enjoyment	0.080	0.027	0.004	0.026	0.133
Progress Feedback → Meaning	0.285	0.035	< .001	0.218	0.353
Meaning → Enjoyment	0.759	0.017	< .001	0.726	0.793
Indirect Effect Clarity of Goals and Rules	0.014	0.006	0.014	0.003	0.026
Indirect Effect Progress Feedback	0.015	0.006	0.008	0.004	0.027
Indirect Effect Meaning	0.062	0.024	0.009	0.015	0.108
Indirect Clarity of Goals and Rules through Meaning	0.022	0.006	< .001	0.009	0.034
Indirect Progress Feedback through Meaning	0.042	0.009	< .001	0.025	0.059
Total Indirect Effect	0.154	0.027	< .001	0.102	0.207
Direct Effect	0.032	0.020	0.114	-0.008	0.072
Total Effect	0.186	0.034	< .001	0.121	0.252

Table 13: Model fit indices for final Philanthropist model

	Baseline test			Fit Indices		
	χ^2	**df**	**p-value**	**CFI**	**RMSEA**	**SRMR**
Final Philanthropist Model	0.000	0	-	1.000	-	-

5. Discussion

In my work, I investigated how specific player experience dimensions shape the relationship between player type score and enjoyment. The final models reveal distinct motivational pathways for Achievers and Philanthropists, offering insight into how different player types experience enjoyment through specific player experience dimensions.

For Achievers, the final model shows that the relationship between player type score and enjoyment is mainly mediated by the experience of *challenge* and *mastery*. Both dimensions were also influenced by *clarity of goals and rules* and *progress feedback*, suggesting a layered structure where some experiences serve as foundations for others. *Challenge* was found to be the strongest predictor of enjoyment (β = .390), followed by mastery (β = .335). These results confirm that Achievers are most engaged when they face meaningful challenges and can develop a sense of competence through overcoming them. Interestingly, *progress feedback* did not directly predict enjoyment, but only when it contributed to *challenge* or *mastery.* This supports the idea that feedback is not inherently motivating on its own, but becomes engaging when it supports a sense of competence or progression. Similarly, while the direct effect of *clarity of goals and rules* on enjoyment was relatively small, it played a key role in shaping the experience of *challenge* and *mastery*. The Achiever model, therefore, highlights how structured feedback and clear objectives can support Achievers in experiencing growth and accomplishment, which ultimately leads to greater enjoyment. Altogether, these findings show that the relationship between Achiever score and enjoyment is partially mediated by increases in *clarity of goals and rules*, *challenge,* and *mastery*.

The final Philanthropist model showed a different motivational pattern. Enjoyment for Philanthropists was almost entirely driven by meaning (β = .759), which, in turn, was shaped by *clarity of goals and rules*, *progress feedback*, and the Philanthropist score. This highlights that Philanthropists are most engaged when they feel their actions serve a purpose or contribute to the greater good. *Clarity of goals and rules* and *progress feedback* did not have strong direct effects on enjoyment directly, but both significantly influenced the experience of *meaning*. However, the strongest effect on *meaning* came from *progress feedback* (β = .285). The model highlights that relevant feedback and clear objectives help Philanthropists perceive their actions as more meaningful. In general, the relationship between Philanthropist score and enjoyment was fully mediated by the PXI dimensions. This means that the effect of a player's Philanthropist score on enjoyment is entirely explained by how much the player perceives meaning, clear goals and rules, and progress feedback in the game.

In both models, sequential mediation paths were observed (e.g., progress feedback → challenge → enjoyment), highlighting that PXI dimensions do not operate in isolation. Instead, they build on one another in meaningful ways. This layered structure suggests that certain experiences only become impactful when supported by others. For example, meaningful feedback may be more engaging when it helps players develop a sense of challenge, and the experience of challenge may be more enjoyable when supported by clear goals, such that players know what different actions they can take. These findings emphasize the importance of designing cohesive experience sequences, rather than relying on isolated gamification features. A purely feature-based approach risks overlooking the underlying mechanisms that truly drive player enjoyment.

In general, the findings of this study align well with earlier research. The Achiever model shows that enjoyment is primarily driven by the experience of *challenge* and *mastery*. This is consistent with how Tondello et al. (2016) describe Achievers: "Achievers are motivated by competence. They seek to progress within a system by completing tasks, or prove themselves by tackling difficult challenges." The strong role of *challenge* and *mastery* in the model reflects this motivation and supports the idea that Achievers enjoy systems that help them grow their skills and overcome obstacles.

Similarly, the Philanthropist model highlights the importance of *meaning* as a key predictor of enjoyment. This aligns with the definition of Tondello et al. (2016) of Philanthropists as players who are "motivated by purpose. They are altruistic and willing to give without expecting a reward." The strong role of *meaning* in the model supports the idea that Philanthropists enjoy experiences that allow them to help others, contribute to the greater good, or engage with a purposeful narrative.

The results showed that PXI dimensions mediate the relationship between player type scores and enjoyment. Interestingly, the moderation analyses were not supported by the data. This suggests that players reach enjoyment through different types of experiences (PXI dimensions), rather than responding differently to the same experiences, as a moderation model would suggest. Earlier research describes player types as motivational profiles that shape how people interact with gamified systems (Krath et al., 2023; Marczewski, 2015; Tondello et al., 2016). Player types seem to guide players towards experiences that matter most to them, rather than changing how strongly a given experience affects enjoyment. For example, Socializers and Philanthropists seek out interaction with other players, while Achievers are more focused on their own performance. This is supported by the strong mediation effects observed in the models: enjoyment increases when players engage with experiences that align with their intrinsic motivations. In other words, different player types are attracted to different game elements, not because the impact of those experiences differs depending on player type.

5.1 Implications for Gamification Design

The findings of this study have several implications for the design of gamified systems. The models show how different player types experience enjoyment through specific player experience dimensions (PXI) and highlight the importance of tailoring gamification to player type profiles. Rather than a one-size-fits-all approach, effective gamification design should be designed around the experience dimensions that matter most to each player type.

The Achiever model shows that enjoyment is mainly experienced through *challenge* and *mastery*, which are both influenced by *clarity of goals and rules* and *progress feedback*. Therefore, players who are highly motivated by achievement benefit from gamified systems that are focused on progression, clear objectives, and opportunities to develop and test their skills over time. Effective gamification elements for these players include visible indicators of progress (e.g., points, badges, XP bars), leaderboards, levels, and challenges that gradually increase in difficulty. These insights align with earlier research, which found that Achievers are especially motivated by elements such as points, levels, challenges, and leaderboards (Krath & von Korflesch, 2021; Tondello et al., 2016).

For Philanthropists, the model shows that enjoyment is strongly driven by a sense of *meaning*, which is influenced by *clarity of goals and rules*, and *progress feedback*. This highlights the importance of incorporating elements that provide a sense of contribution or purpose, even in environments without strong social features. This is in line with previous research that has shown that Philanthropists prefer elements that give them a sense of meaning such as gifting, knowledge sharing, and administrative roles (Krath & von Korflesch, 2021; Tondello et al., 2016). In addition to these elements, I suggest that designers include elements such as storylines or missions that are focused on helping others (even if they are non-playable characters), or symbolic feedback in environments where social interaction is limited or absent. These elements can evoke a sense of meaning, even without explicit collaboration or multiplayer features. Symbolic feedback uses symbols to communicate information, such as a tree growing, or a heart icon filling up.

This study uses the player types Achiever and Philanthropist as the basis for exploring player experiences. The models highlight how different player types experience enjoyment through different psychological pathways, showing that the same game element can have very different effects depending on who is using it. For example, while both Achievers and Philanthropists benefit from progress feedback, the way it supports their enjoyment depends on whether it enhances a sense of mastery or meaning. This suggests that player types are a useful lens not just for selecting elements, but also for understanding how to shape deeper experiences. Gamification designers can use player types to build experience sequences that match specific motivational profiles. In this way, player types serve as a useful bridge between design decisions and the underlying psychological needs of players.

While much of the literature on gamification has focused on matching player types to individual game elements, this study emphasizes the interdependencies between experience dimensions. In both models, sequential mediation paths were found (e.g., progress feedback → mastery → enjoyment), suggesting that PXI dimensions do not operate in isolation but instead build on each other. This has important implications for design. Rather than focusing on individual motivators like badges, points, or leaderboards, designers should adopt a layered design strategy. This means considering how features interact to build toward a deeper, more cohesive player experience. For example, instead of merely implementing a level system, designers could frame levels within a larger mission structure that supports skill development and personal progression, for example with storylines, missions, or symbolic feedback. This perspective calls for a shift from feature-based design to experience-based design. Instead of thinking in terms of "adding points" or "including a leaderboard," designers should ask how each element contributes to a meaningful player experience. For example, for Philanthropists, progress feedback is most effective when it enhances a player's sense of meaning, not just when it reports progress. Designing with these experience sequences in mind may lead to more engaging gamified systems, as they better reflect how different player types experience enjoyment through different PXI dimensions.

Moreover, the Achiever model showed that *progress feedback* had no direct effect on enjoyment, but only indirect effects via *challenge* and *mastery*. This supports the idea that feedback is not intrinsically motivating in itself, but it becomes meaningful when it supports a deeper sense of competence or progression. Therefore, designers should think carefully about how feedback can enhance perceived challenge and mastery. Designers should therefore carefully consider how feedback enhances perceived challenge or mastery. Instead of simply displaying scores or badges, systems might use dynamic, personalized feedback like "You solved this 20% faster than last time" or "You've reached a new skill tier," which directly reinforces the player's sense of progression.

Finally, these findings also have implications for Games User Research (GUR). Researchers should pay attention to the layered structure of player experiences and study how combinations of features influence sequences of psychological responses. While simple measures of feature preference are helpful, they may not capture the full picture. More insights can be gained by exploring how different elements work together over time to create engaging experiences. Experimental and longitudinal methods could be especially valuable for identifying the pathways through which design elements lead to enjoyment for different player types.

In summary, designing effective gamified systems requires more than aligning elements with player types. It involves creating interconnected experiences that reflect the complexity of human motivation. By understanding how different elements interact and build on each other, designers can create more meaningful and engaging systems.

5.2 Limitations and Future Work

One important limitation concerns the interpretation of the PXI data. The PXI measures how strongly players experienced specific dimensions of a game or gamified system (e.g., *challenge, clarity of goals and rules, progress feedback*), but it does not capture how much they value these dimensions. As such, higher scores indicate a greater perceived presence of a particular experience, not necessarily its subjective importance to the player. In this study, assumptions were made about which PXI dimensions would be most relevant for Achievers and Philanthropists, and these dimensions were included in the models. However, due to the nature of the dataset, it was not possible to directly test whether players who score high on these player types actually prioritize these dimensions over others. For example, the Achiever model shows that higher experiences of challenge are associated with higher enjoyment, but this does not imply that Achievers find challenge the most important aspect of their experience. Future research could try to directly test which PXI dimensions are prioritized by each player type. If different dimensions are found to be important, a better theoretical model can be developed afterward.

Additionally, it is important to mention that the selection of relevant PXI dimensions for the Achiever and Philanthropist priles was based on my own interpretation of existing psychological theories, such as the Self-Determination Theory (Deci & Ryan, 1985) and the Four Drives Theory (Furneaux & Rieser, 2022). However, these interpretations have not been empirically validated in the context of the Hexad model. To the best of my knowledge, there is currently no established research directly linking Hexad player types to specific psychological theories or PXI dimensions. Therefore, while the selected dimensions were grounded in theory, they should not be interpreted as a confirmed mapping between player types and PXI dimensions. It would be very useful to investigate these hypotheses and study how Hexad player types relate to psychological theories to get a better understanding of the underlying motivations of each player type.

Another limitation concerns the nature of Likert-scaled data, which can introduce biases. All variables used in this thesis were measured through 7-point Likert scales. Individual differences in response styles can influence how people use these scales as some participants tend to answer more extremely, while others are more moderate in their responses. This variability in scale use can affect the comparability of scores across individuals and may introduce measurement bias. Additionally, Likert scales rely on self-reported data, which introduces the possibility of biases such as the social desirability. While self-reporting is common in experience-based research, future studies could try to combine subjective reports with behavioral data (e.g., in-game actions or completion rates) to gain a more objective view of player experiences.

Next, there are limitations related to the inclusion of the Philanthropist player type in the scope of this study, while Tunnel Runner does not clearly incorporate social elements. Philanthropists value opportunities for social interaction, cooperation, or helping others. While some players may have experienced a minimal sense of connection through the presence of the rats, the game did not offer actual opportunities to engage with or support others. These missing features are central to the motivations of Philanthropist player type. The limited social context may have influenced their experience of the game, and consequently their reported enjoyment. As the model selection process was purely data-driven, it is possible that some players with high

Philanthropist scores reported relatively low enjoyment, simply because the game might not have aligned well with their core motivational preferences. This could have influenced the strength or direction of the relationships in the Philanthropist model.

The models developed in this study are based on experiences within the Tunnel Runner game, which restricts the generalizability of the findings. Game design choices such as goal, structure, and aesthetics all influence how players experience the game, and thus the scores of the PXI dimensions. It is possible that the same player types would experience enjoyment differently in a game with a different genre, goal, or design. Moreover, the models in this thesis have been developed and tested on a single dataset and have not yet been validated with another sample. While the models showed clear pathways and meaningful relationships within this dataset, this needs to be validated more often to be able to generalize the results. Without replication or cross-validation, it can not be proven whether the identified relationships would hold in different game contexts, or across different samples. Future research should try to test these models on new datasets to evaluate their broader applicability.

Furthermore, the model selection process is a subjective process. Although all relationships in the model were guided by theory, the choices about which paths to include or exclude based on modification indices ultimately involved my own interpretation and judgment. Different researchers might have made slightly different decisions, and this could have led to alternative model structures. As such, the presented models should be viewed as one possible representation of the underlying relationships, rather than a final explanation. In the future, the model selection process could be performed by multiple researchers to increase reliability of the models.

Although the Philanthropist model showed perfect fit indices, it is important to be critical of the interpretation. The perfect fit arises because the model is just-identified, leaving no degrees of freedom. As a result, the fit indices are mathematically guaranteed to appear perfect, but do not reflect how well the model captures the underlying structure of the data. Perfect fit may suggest overfitting or that the model is too closely related to the data sample. However, in this study, the model structure was relatively simple and the pathways were both theoretically plausible and supported by the data. It can therefore still be seen as reliable, but it is important that future research should try to test the robustness and generalizability of this model using different datasets or experimental conditions.

Lastly, this study focused only on two player types from the Hexad framework, Achievers and Philanthropists. As such, the models do not provide insights into the motivational relationships of other player types, such as Disruptors, Free Spirits, Players, or Socializers. Future research should expand the scope to include these types to create a more comprehensive picture of how different player types interact with PXI dimensions.

6. Conclusion

In conclusion, this thesis investigated how different PXI dimensions can explain the relationship between Achiever and Philanthropist scores, and enjoyment in a gamified system. Model analyses revealed that player experience dimensions mediate this relationship and operate through sequential pathways. The findings reveal that each player type finds enjoyment through different factors. For Achievers, enjoyment is primarily driven by the dimensions *challenge* and *mastery*, which are influenced by *clarity of goals and rules*, and *progress feedback*. In contrast, Philanthropists experience enjoyment mainly through *meaning*, which is shaped by *progress feedback* and *clarity of goals and rules* as well. These insights confirm that enjoyment in gamified systems is not just the result of individual elements or player preferences, but of a sequence of related experiences that develop throughout gameplay. Therefore, designers should not only focus on isolated gamification features, but also consider how different elements interact to form cohesive and meaningful experiences. By designing for interconnected experiences, gamified systems maximize their potential to engage different users and create enjoyable experiences tailored to players' individual preferences.

7. The use of AI

During this thesis, I made use of the ChatGPT (OpenAI, 2023). I made sure to follow the Eindhoven University of Technology guidelines for AI use. ChatGPT was mostly used to refine language, restructure complex paragraphs, and improve the clarity and coherence of the text. In some cases, it served as a sparring partner to help explore alternative perspectives, for example when discussing interpretations or practical implications. All content and scientific reasoning were developed independently by me. ChatGPT was used solely as a writing assistant to support the communication of my ideas, not to generate or interpret research data, perform statistical analyses, or draw conclusions.

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Appendices

Appendix A: Hexad-12 questionnaire

Table 14: 12-item short version of the Hexad scale (Hexad-12)

Hexad Type	Item	Question
Philanthropist	Phi1	It makes me happy if I am able to help others.
	Phi2	The well-being of others is important to me.
Socializer	Soc2	I like being part of a team.
	Soc4	I enjoy group activities.
Achiever	Ach2	I like mastering difficult tasks.
	Ach4	I enjoy emerging victorious out of difficult circumstances.
Player	Pla2	Rewards are a great way to motivate me.
	Pla4	If the reward is sufficient, I will put in the effort.
Free Spirit	Fre1	It is important to me to follow my own path.
	Fre3	Being independent is important to me.
Disruptor	Dis3	I see myself as a rebel.
	Dis4	I dislike following rules.

Appendix B: PXI questionnaire

Table 15: PXI dimensions and items

PXI dimension	Item	Question
Autonomy	Autonomy_1	I felt free to play the game in my own way.
	Autonomy_2	I felt like I had choices regarding how I wanted to play this game.
	Autonomy_3	I felt a sense of freedom about how I wanted to play this game.
Curiosity	Curiosity_1	I wanted to explore how the game evolved.
	Curiosity_2	I wanted to find out how the game progressed.
	Curiosity_3	I felt eager to discover how the game continued.
Meaning	Meaning_1	Playing the game was meaningful to me.
	Meaning_2	The game felt relevant to me.
	Meaning_3	Playing this game was valuable to me.
Mastery	Mastery_1	I felt I was good at playing this game.
	Mastery_2	I felt capable while playing the game.
	Mastery_3	I felt a sense of mastery playing this game.
Immersion	Immersion_1	I was no longer aware of my surroundings while I was playing.
	Immersion_2	I was immersed in the game.
	Immersion_3	I was fully focused on the game.
Feedback	Feedback_1	The game informed me of my progress in the game.
	Feedback_2	I could easily assess how I was performing in the game.
	Feedback_3	The game gave clear feedback on my progress towards the goals.
Challenge	Challenge_1	The game was not too easy and not too hard to play.
	Challenge_2	The game was challenging but not too challenging.
	Challenge_3	The challenges in the game were at the right level of difficulty for me.
Control	Control_1	It was easy to know how to perform actions in the game.
	Control_2	The actions to control the game were clear to me.
	Control_3	I thought the game was easy to control.
Goals/Rules	Goals_1	I grasped the overall goal of the game.
	Goals_2	The goals of the game were clear to me.
	Goals_3	I understood the objectives of the game.
Enjoyment	Enjoyment_1	I liked playing the game.
	Enjoyment_2	The game was entertaining.
	Enjoyment_3	I had a good time playing this game.